

Cross-asset lead-lag into euro government-bond futures

Do US rates, crude and equity futures lead the Eurex govvie curve — and is it tradeable?

Results — 1-minute bars & 1-second order book · July 2026

Data & method

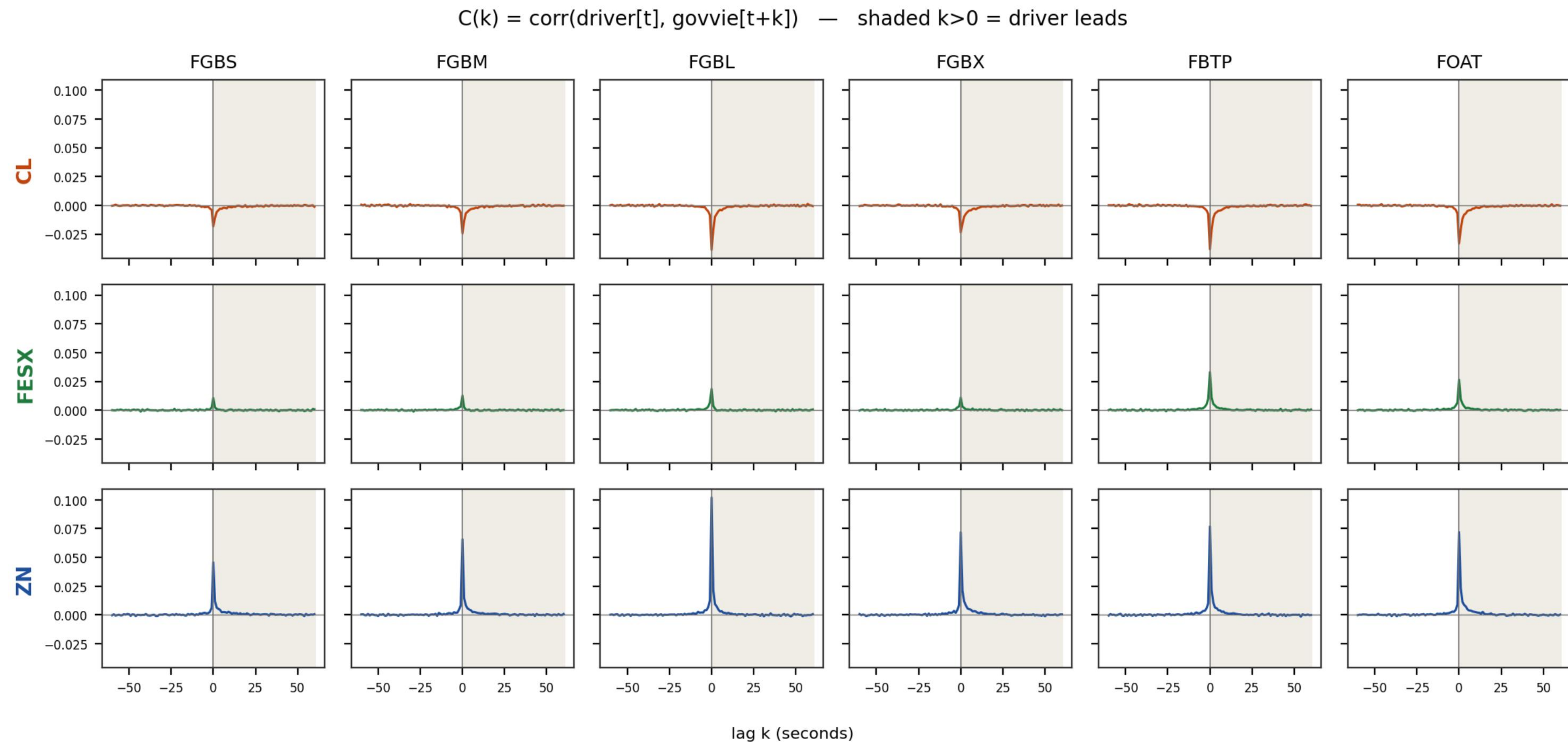
- **Instruments.** Eurex govies — Schatz, Bobl, Bund, Buxl, BTP, OAT. Drivers — ZN (US 10Y), CL (WTI crude), FESX / ES (equity).
- **Data.** Databento TBBO top-of-book (1 s mids) and OHLCV 1-minute bars. Session 08:00–17:00 Europe/Berlin. 2025-03 → 2026-06; 231 common days at 1 s.
- **Metric.** $C(k) = \text{corr}(\text{driver return}[t], \text{govvie return}[t+k])$. $k=0$ is contemporaneous; $k>0$ means the driver leads by k . Log-returns, winsorized 0.1 / 99.9%.
- **Tradeable leads only.** Strictly-leading lags ($k \geq 1$); lag 0 is same-bar co-movement and not actionable.

1-minute lead-lag correlation, computed directly

driver → govvie (lead k = 1 min)	mean corr	mean signed	% corr > 2 σ
ZN → Bund	0.085	−0.000	8.5%
ZN → BTP	0.084	+0.002	8.5%
ZN → OAT	0.082	+0.006	7.9%
CL → BTP	0.078	−0.010	6.6%
CL → Bund	0.074	−0.005	5.4%
noise (unrelated series)	0.073	0.000	~5%

Averaged directly over all ~9,100 rolling windows per pair — no train/test split. The 1-minute lead correlation is $\approx \mathbf{0.08}$, right at the $\approx \mathbf{0.073}$ chance level, and $\approx \mathbf{0}$ signed, for every pair including BTP. There is no persistent 1-minute lead; the +0.045 pooled BTP figure comes from a few volatile episodes, not a repeatable edge.

1-second cross-correlation, all pairs (231 days)



The cross-correlation is **asymmetric**: ZN and CL carry more weight on the $k > 0$ side (driver leads) than $k < 0$. FESX is symmetric — synchronous, not leading. The lead is concentrated within ~1 second.

1-second: contemporaneous vs peak lead ($k > 0$)

driver → govvie	k = 0	peak lead	lag
ZN → Bund	+0.102	+0.022	+1 s
ZN → BTP	+0.077	+0.022	+1 s
ZN → OAT	+0.072	+0.023	+1 s
CL → Bund	−0.039	−0.021	+1 s
CL → BTP	−0.038	−0.021	+1 s
FESX → BTP	+0.033	+0.011	+1 s

A genuine lead exists at 1 second (peak at +1 s), an order of magnitude larger than at 1 minute. ZN leads across the whole curve; FESX barely.

ZN → Bund: correlation by lag

lag	correlation
same second (k = 0)	0.102
1-second lead	0.022
1-minute lead	0.003

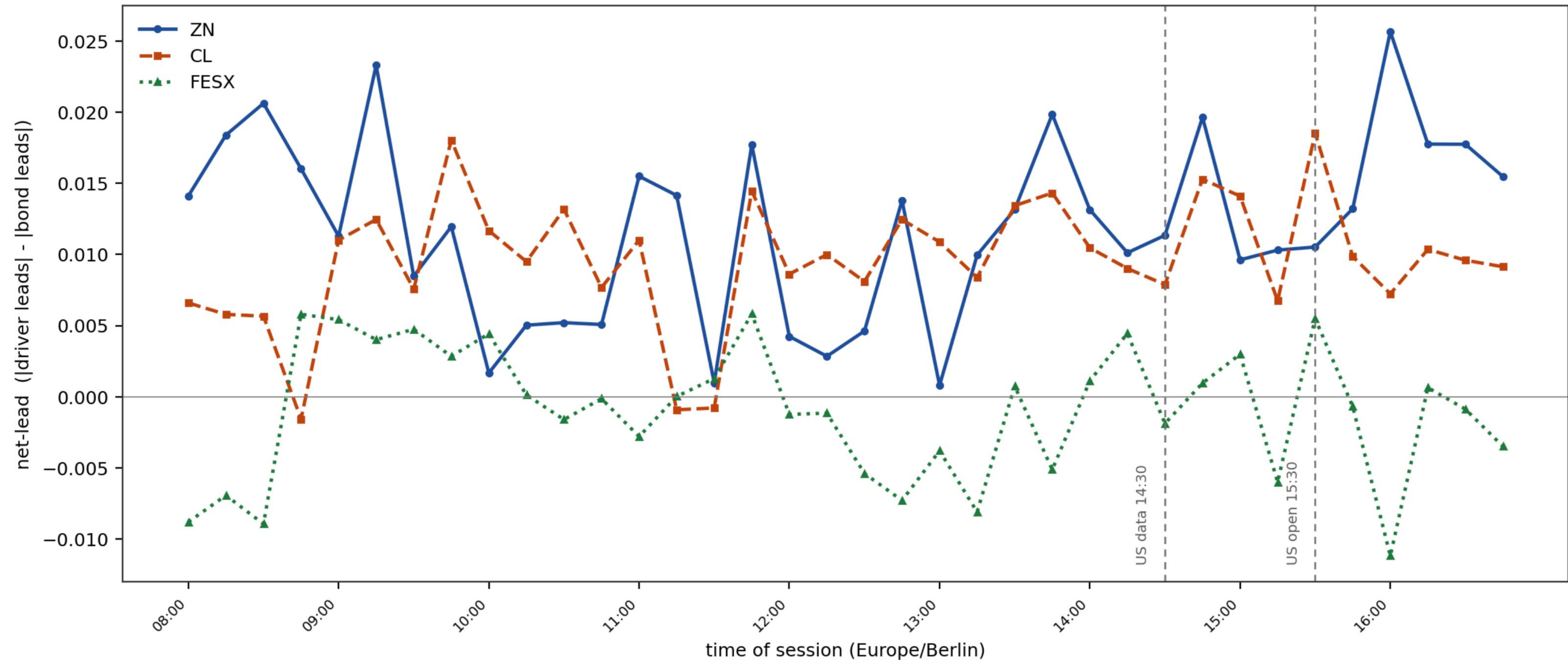
The relationship is mostly instantaneous. The lead decays within seconds and is gone by one minute — a sub-second microstructure effect, not a bar-scale signal.

Is the 1-second lead tradeable?

quantity	value
lead correlation	≈ 0.02
variance explained (R^2)	0.04%
directional hit-rate (sign)	50.2%
follow-through, top-decile ZN move	+0.12 ticks
BTP 1-minute move (std)	3.34 ticks
round-trip cost (spread + fee)	$\sim 1\text{--}2$ ticks

Statistically certain, economically negligible: the follow-through is far below the round-trip cost and direction is a coin flip. **Not tradeable as a taker signal.**

When does the lead occur? Net-lead by time of session



ZN leads the govies all session and strengthens into the US cash open (peak 16:00; US hours 1.35× the morning). CL is weaker (1.18×). FESX does not lead at any time.

Does ZN improve a TLOB model of BTP?

metric	BTP book only	BTP + ZN
test accuracy	0.6428	0.6413
accuracy (thresholded)	0.6502	0.6506
F1	0.335	0.343
test loss	0.867	0.871
val loss (best)	0.909	0.909

Stacking ZN's order book onto BTP's (9 → 17 features) changes nothing beyond run-to-run noise. BTP's own book already contains the cross-asset information.

Summary

- ZN (US 10Y) and CL (WTI) lead the euro govies; FESX / equity is synchronous, not leading.
- The relationship is contemporaneous-dominated. A genuine lead exists only at ~1 second and has decayed by one minute.
- Magnitude is tiny: lead correlation ≈ 0.02 (R^2 0.04%), sign accuracy $\approx 50\%$ — below transaction cost.
- The lead concentrates in the US session (ZN peaks ~16:00, 1.35× the morning).
- Adding ZN as a feature does not improve a TLOB directional model of BTP.